

Omkar Honrao

Robotics Research intern | Python | Autonomous System

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GitHub : <https://github.com/omkar-robotics>

Education

- **Bachelor of Technology in Electrical engineering** 2025-28
Zeal college of engineering and Research Narhe ,Pune
- **Diploma in Electrical Engineering** 2023-25
Government Polytechnic, Nandurbar
- **ITI ELECTRICIAN** 2019-22
Gramin technical and management campus Nanded

Personal Projects

- **ROS2-Based Autonomous Mobile Robot Navigation using SLAM & Nav2**
 - Implemented autonomous robot navigation using SLAM Toolbox for environment mapping and Navigation2 for path planning
 - Simulated robot movement in Gazebo and visualized mapping, localization, and robot trajectory in RViz2
 - Configured robot localization using AMCL and tested autonomous navigation in simulated environments
 - **Technologies Used** : ROS2 Humble, SLAM Toolbox, Navigation2, Gazebo, RViz2
- **Hand-Following Robot – Computer Vision Pipeline**
 - Developed real-time hand gesture tracking system using OpenCV and MediaPipe
 - Detected and tracked 21 hand landmarks to control robot movement based on hand position
 - Implemented gesture-based control logic for robot navigation commands (Forward, Left, Right, Stop)
 - **Technologies Used**: Python, OpenCV, MediaPipe
- **ROS2 Action Client-Server System**
 - Developed a custom ROS2 Action client-server system using Python and ROS2 Humble
 - Implemented asynchronous goal handling, continuous feedback, result processing, and goal cancellation
 - Created a custom CountUntil.action interface for long-running robotic task execution
 - **Technologies Used**: ROS2 Humble, Python, Custom Actions

Experiences

- **Robotry — Robotics Research Intern** April - 2026 – Present
 - Working on ROS2-based robotics system development and robot communication
 - Developing publisher-subscriber, service, parameter, and action-based ROS2 applications using Python
 - Learning robotic software architecture, Linux environment, and autonomous system concepts
 - Contributing to robotics research and practical implementation tasks
- **Tata Motor Passenger Vehicles Ltd . - Industrial Trainee** March - June 2022
 - Worked on industrial electrical systems in automobile manufacturing environment
 - Assisted in machine wiring, control systems, and industrial equipment setup
- **Aditya Power (Government Certified Solar & Electrical Engineer)** June- July 2024
 - Received hands-on exposure at 220/132 kV Substation & Solar Power Generation Plant
 - Learned basics of power generation, transmission, and substation equipment
 - Observed operation of transformers, switchgear, protection systems, and control panels

TECHNICAL SKILLS and INTERESTS

Programming : Python

Robotics : ROS2 Humble & Jazzy , Topics, services, Actions, Parameters, SLAM, Navigation, AMCL

Simulation & visualization : Gazebo , Rviz2

CAD & Design : Fusion 360

Operating System & Tools : Linux, Git/GitHub

Computer vision : OpenCV , MediaPipe

Areas of Interest:

Robotics & AI, ROS 2 Development, Autonomous system, Computer Vision

Soft skill:

Public Speaking, Anchoring, Effective Communication,
Stage Confidence, Team Coordination, Adaptability

POSITIONS of RESPONSIBILITY

- Event Anchor / Host — College Events
- Anchored multiple technical and cultural programs